

Employment

Graduate Research Assistant NC State University Apr 2025 – Present

- Collaborated with plant biologists and geneticists to deploy cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling 100+ researchers to visualize 500GB+ genome datasets.
- Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.

Product Engineer Intern Isomer AI May 2025 – May 2026

- Collaborated with product and design teams to ship 10+ production AI features including Auth0 RBAC access control and multi-model LLM pipelines (Claude, GPT-4o, Gemini); 923 commits, 7 production deployments.
- Owned full-stack React/Node.js + TypeScript codebase: PostgreSQL, Redis, Docker, Auth0 RBAC, and Playwright test coverage.

Software Developer Intern Mphasis (HP Company) Jun 2024 – Aug 2024

- Partnered with healthcare data teams to build 5+ AI workflows including health claim automation using OCR, NLP, and ML pipelines; improved document processing accuracy by 25% across 10K+ daily medical claims.
- Automated AWS and GCP CI/CD pipelines; integrated Grafana monitoring and led model validation and red-teaming sessions.

AI/ML Intern Capgemini Jun 2023 – Aug 2023

- Led cross-functional SDLC for data-driven enterprise ticket analysis using NLP classification models; worked with support teams to reduce resolution time by 15% and eliminate 2,000+ monthly manual triages.
- Built Kafka + Spark + Hadoop pipelines with data engineers; cut ticket-processing latency by 20% at peak load.

Education

Raleigh, NC North Carolina State University Aug 2024 – May 2026

- Master of Science in Computer Science
- Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.

Mumbai, India Mumbai University Aug 2020 – May 2024

- Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.

Projects

CortexQ Kubernetes LLM Inference Platform 2026

- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing.
- Deployed via Helm + ArgoCD GitOps on Kubernetes; OTel + Prometheus + Grafana observability; 88 pytest tests.

PulseRank Real-Time Leaderboard & Activity Feed 2026

- Built event-driven leaderboard and activity feed in TypeScript with NestJS and Next.js: MySQL transactional writes, MongoDB feed projections with TTL, Redis sorted sets for sub-millisecond reads; live SSE.
- Validated multi-DB consistency with Testcontainers; benchmarked hot paths with k6 at 10K concurrent users.

LLM Eval Harness Independent Project 2025

- Built vendor-neutral LLM evaluation harness in Python comparing Claude, GPT-4o, and Gemini across 2,000 frozen labeled examples spanning factual recall, reasoning, and code generation tasks.
- Tracked hallucination rate, p50/p95 latency, and cost-per-call; applied McNemar significance tests across 6 metrics.

Languages and Technologies

- **Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash
- **Identity & Security:** Auth0, RBAC, OAuth, REST APIs, Docker, Kubernetes, Terraform
- **Cloud & Infra:** AWS, GCP, FastAPI, Node.js, NestJS, PostgreSQL, Redis, MongoDB, MySQL
- **Observability & CI/CD:** OpenTelemetry, Prometheus, Grafana, Sentry, GitHub Actions, Helm, ArgoCD

Additional Experience and Awards

- **IEEE Published:** Co-authored and presented research on Computer Vision-Based Cybersecurity Threat Detection at the Springer icSoftComp 2023 international conference, reaching an audience of 500+ global attendees.
- **Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; built 3 full-stack MVPs end-to-end in 48-hour competitive sprints.
- **GDSC Lead:** Led 4+ tech events as Google Developer Student Club Lead, reaching 500+ students across hackathons and conferences.
- **Technical Writing:** Published articles on LLM model collapse, AI agent security, and audio AI at kulkarniaditya.com/blog.